Field Hours

CSCE 650 Virtual Reality

Original notes for this site. Same topics as the catalog course. Not a reprint of another university's lectures, and not academic credit.

Lecture 1 · 90 minutes

A field of view is not presence

You separate immersion from presence before you judge a headset. Immersion is what the device supplies: field of view, tracking, and stereo. This exercise uses a headset with a 100 degree horizontal field and tracking of the head plus 2 controllers. Presence is the sense of being in the place you drew. You can have the 100 degree field and still have no presence if the world does not answer when the head turns.

You test the split with one action. The user turns their head 30 degrees left. The view must rotate by those same 30 degrees. A delay past about 20 milliseconds is a failed coupling you can feel. That coupling is immersion. Whether the user then ducks a virtual beam is evidence about presence. You do not score the duck from the field-of-view number.

Immersion is the device coupling. Presence is what the user does with it.

Immersion

A 100 degree field

Head plus 2 controllers

Delay near 20 ms

Presence

Sense of being there

Not a device spec

Read from behavior

A 30 degree head turn

1. The headset reports a yaw change of 30 degrees.

2. You rotate the view by 30 degrees. If the image trails by more than about 20 milliseconds, the coupling has failed.

3. You do not mark the user present because the field is 100 degrees. You watch whether they duck the beam.

Problem 1. The headset has a 100 degree field and the view ignores a head turn. Which side, immersion or presence, has already failed?

Answer. Immersion. Tracking that does not rotate the view by the reported angle fails the device coupling. You cannot read presence from a view that did not follow.

Problem 2. The view matches a 30 degree turn well under 20 milliseconds, and the user does not react to a virtual beam. What may you conclude?

Answer. The immersion coupling worked. You may not conclude presence. Presence needs the behavioral evidence, and here it was absent.

Lecture 2 · 90 minutes

A floor and an eye height

You match the virtual floor to the real floor at one mark, the origin. Eye height here is 1.6 meters, so the camera starts at y = 1.6. At y = 1.2 the floor reads 0.4 meters too close. You check with a real 1 meter stick on the mark. It should meet a virtual segment from y = 0 to y = 1. The stick is the calibration mechanism.

You track the 2 controllers in the same frame as the head. A controller at (0.3, 1.0, 0.4) is 0.3 meters right of the origin, 1.0 meters up, and 0.4 meters forward. You do not give the hand its own origin. A separate origin can put a hand that is beside the head in the room a meter away in the world, and a grab fails. You recompute the controller from the same tracker that applied the 30 degree yaw.

One origin serves the floor, the eyes, and both controllers.

Origin on the real floor

Eyes at 1.6 meters

Controllers share that frame

Check with a 1 m stick

The stick on the mark

1. The origin is a mark on the real floor. The camera starts at y = 1.6.

2. A 1 meter stick on the mark should line up with a virtual segment from y = 0 to y = 1.

3. A controller at (0.3, 1.0, 0.4) uses that same origin. A second origin for the hand would break the grab.

Problem 1. You set the camera at y = 1.2 and the eye height is 1.6 meters. By how much is the virtual floor wrong?

Answer. 0.4 meters too close. The camera is 0.4 meters below the eyes, so the floor appears nearer than the real floor.

Problem 2. Why must a controller use the head origin?

Answer. So a hand beside the head in the room stays beside the head in the world. A separate origin can place the hand a meter away, and the grab fails.

Lecture 3 · 90 minutes

A ray and a button

You select with a ray that starts at the controller, here (0.3, 1.0, 0.4), and runs along the controller forward axis.

The first object hit within 3 meters is the candidate. You confirm with a button press, not with the hit alone, so a ray that sweeps a menu does not select every icon it crosses. A hit at 4 meters is outside the range and is not a candidate, even if the button goes down.

You carry a selected object by storing the offset at the press. If the hit was 1 meter along the ray, you keep that 1 meter offset while the button is held. A controller motion of 0.2 meters moves the object by 0.2 meters. Leaving the object at the original hit point would not let the user carry it. Releasing the button drops the attachment and keeps the last pose. The object does not snap back.

The ray proposes. The button confirms. The stored offset does the carrying.

Ray from the controller -> First hit within 3 m -> Button confirms -> Offset held while down

Carry by 0.2 meters

1. The ray starts at (0.3, 1.0, 0.4). The first hit is 1 meter out, inside the 3 meter range.

2. The button stores that 1 meter offset and attaches the object to the controller.

3. A later controller motion of 0.2 meters moves the object by 0.2 meters. Release keeps the new pose.

Problem 1. The first hit is 4 meters out and the user presses the button. What is selected?

Answer. Nothing. The range is 3 meters, so a hit at 4 meters is not a candidate. The button does not override the range.

Problem 2. The object is attached at an offset of 1 meter, and the controller then moves 0.2 meters. How far does the object move?

Answer. 0.2 meters. The offset stays fixed in the controller frame, so the object follows. On release it keeps the new pose.

Lecture 4 · 90 minutes

Walking and teleport

You do not map the gallery onto the real floor one meter for one meter. The tracked space is 2 meters by 2 meters. The gallery is 10 meters long, so walking covers only 2 meters. You add a teleport: the controller ray aims at the floor, a second button moves the origin, and the feet land there. You keep head yaw. Resetting yaw makes the user relearn where the real door is.

You reject a target that would put the body in a wall. A floor point at least 0.5 meters from any wall is legal. A point 0.2 meters from a wall is not, because the body occupies space the ray did not measure. You also reject a teleport while an object is attached, so a carry and a jump are not one press. Two buttons, two actions. Between teleports the user can still turn inside the 2 meter space.

Walking uses the tracked floor. Teleport moves the origin and keeps yaw.

Real walking

Space is 2 by 2 m

Feet move the camera

Stops at the boundary

Teleport

Gallery is 10 meters

Ray aims at the floor

Yaw stays as it was

A legal jump down the gallery

1. The user stands in a 2 by 2 meter space and aims at a floor point 6 meters along the 10 meter gallery.

2. The point is 0.5 meters from the nearest wall, so you move the origin and you do not change yaw.

3. A point 0.2 meters from a wall is rejected. So is any target while an object is still attached.

Problem 1. The tracked space is 2 meters by 2 meters and the gallery is 10 meters long. Why is walking not enough?

Answer. Walking covers only the 2 meter space. The rest of the 10 meter gallery would leave the tracked area. Teleport covers that difference.

Problem 2. A floor target is 0.2 meters from a wall and nothing is attached. Do you teleport?

Answer. No. The clearance rule requires 0.5 meters. Targets you do accept also keep the current yaw.

Lecture 5 · 90 minutes

What breaks the place

You treat presence as easy to spoil with one inconsistency. The user has ducked a beam, which is a useful sign. You then let the hand pass 10 centimeters into the grabbed object because the stored offset was stale. That intersection tells the user the object is a picture. You would rather drop the attachment than draw the hand inside the mesh. The 100 degree field does not repair a hand that occupies the same space as the prop.

You also watch body cues you did not script. If all 4 of your test users lean when the gallery floor appears to drop, the floor cue is working. If none lean, you do not add a louder sound and rename it presence. You change the floor cue and test again. A self-report comes after the behavior. A score of 5 out of 7 from a user who walked through a wall is not the number you lead with.

Behavior is the evidence. A score comes after you have described what people did.

Watch the body first

Hand must miss the mesh

A lean can be evidence

A score of 5 comes later

A hand inside the prop

1. The user selected the object and you stored a 1 meter offset.

2. A later frame draws the hand 10 centimeters inside the mesh because that offset was stale.

3. You drop the attachment rather than show the intersection. The earlier duck of the beam does not cover this frame.

Problem 1. All 4 users walk through a virtual wall, and each reports presence as 5 out of 7. What do you report first?

Answer. That they walked through the wall. The behavior says the wall did not act as a boundary. The score of 5 comes after that fact, not ahead of it.

Problem 2. The hand intersects the prop by 10 centimeters. What do you do with the attachment?

Answer. Drop it. Drawing the hand inside the mesh says the object is not solid. The field of view does not cover that error.

Lecture 6 · 90 minutes

A task and four users

You evaluate with a task and a count, not with a tour. The task is to select 3 objects and teleport to the far end of the 10 meter gallery. You time it and you count errors: a selection past 3 meters, a teleport you accepted inside 0.5 meters of a wall, a dropped object. Four users make a pilot. You do not average 4 people into a general claim. You list every error beside the task that produced it.

You compare two interfaces only when the task stays fixed. Interface A uses the ray. Interface B uses a touch range of 0.4 meters and no ray. If A finishes with 1 error and B with 5, B failed this task in this pilot. You may not say a ray is better in every room. The 4 users and the 2 meter space are part of the claim you write down.

Fix the task, count the errors, and keep the claim inside that pilot.

1. Fix the task in writing

2. Count each error

3. Four users is a pilot

4. Do not widen the claim

Two interfaces, one task

1. The task is 3 selections plus one teleport to the end of the 10 meter gallery.

2. Interface A records 1 error. Interface B records 5. Both counts come from the same 4 users.

3. You report the task and the pilot size with the counts. You do not generalize past this room.

Problem 1. Interface A has 1 error and interface B has 5, on the same task with the same 4 users. What may you claim?

Answer. On this task, in this pilot, B produced more errors than A. You may not claim that a ray is

better for every room and every user.

Problem 2. Why is an unstructured tour of the gallery a weak evaluation?

Answer. A tour has no fixed task and no error count. The assigned task, 3 selections and a teleport, gives times and errors you can list.

Lecture 7 · 90 minutes

A gallery small enough to test

You build one room, not a city. The gallery is 10 meters long, with 3 objects, a floor matched at eye height 1.6 meters, and the 2 meter tracked space at one end. You place the 3 objects inside the 3 meter ray range, so the first selection needs no teleport. The far target sits 0.5 meters clear of the end wall. That room is the whole pilot.

You implement in an order you can test. First the 30 degree yaw, then the 1 meter stick, then the ray and its button, then teleport on the second button, then a log. If the mesh comes first, a failed grab might be the ray, the origin, or the mesh. You log hit distance on every press. A hit at 4 meters is an out-of-range error.

Each piece is testable before the next one depends on it.

Yaw, then the floor -> Ray and its button -> Teleport on button two -> Log every error

The pilot room

1. The room is 10 meters long. The tracked space is 2 by 2 meters at one end. Eye height is 1.6 meters.

2. Three objects sit within 3 meters of the start. The far target is 0.5 meters clear of the end wall.

3. You log hit distance on each press. A hit at 4 meters is the out-of-range error from the selection rule.

Problem 1. Why do you place the 3 objects within 3 meters of the start pose?

Answer. So the first selection can succeed with the ray alone and does not depend on teleport. A failure is then about selection, not about navigation.

Problem 2. A button press logs a hit at 4 meters. What error is that?

Answer. An out-of-range selection. The ray range is 3 meters, so the press must not select that hit. The log records the distance rather than a generic failure.

Lecture 8 · 90 minutes

Run the pilot again

You can now set a headset with a 100 degree field, match a floor at eye height 1.6 meters, select with a ray out to 3 meters, carry an object by a stored offset, and teleport across a 10 meter gallery from a 2 meter play space. You can tell immersion from presence, and you can run a 4-person pilot whose claim stays inside its task. That is a small experience you can defend without adding a second room.

You close by running that pilot and writing behavior plus count. If a user walks through the end wall, the 0.5 meter clearance or the wall itself failed, and a high score does not cancel the walk. If selections fail only while an

object is attached, look at the rule that blocks teleport during a carry. You fix the mechanism the errors name, then you repeat the same task. You do not add a room to escape a bad count.

The same task, a named fix, then the task again. Do not grow the room.

1. Run the same task

2. List behavior and errors

3. Fix the named rule

4. Repeat the pilot

A pilot that met the wall

1. The task is still 3 selections and one teleport in the 10 meter gallery, with 4 users.

2. One user reaches a point 0.2 meters from the wall because you accepted it, and they intersect the wall.

3. You restore the 0.5 meter rejection and run the same task. You do not lengthen the gallery and call the result new.

Problem 1. Combine selection range and presence. A press on a hit at 4 meters selects it anyway, and the hand then passes through the object. What two failures do you write?

Answer. The 3 meter range was not enforced, which is an interaction failure on the immersion side, and the hand intersecting the object is a presence failure because the object no longer behaves as a solid. The 100 degree field offsets neither.

Problem 2. Combine navigation and the pilot. One of 4 users teleports to a point 0.2 meters from a wall. How do you report it, and what do you refuse?

Answer. You count an error on this task: a target inside the 0.5 meter clearance was accepted. You refuse to treat a presence score as the result, and you refuse a claim beyond these 4 users and this task. You fix the rejection and repeat the task.